AN INTERNATIONAL AWARD-WINNING INSTITUTION FOR SUSTAINABILITY

KULLIYYAH OF ENGINEERING
DEPARTMENT OF MECHATRONICS ENGINEERING

PROJECT REPORT

PODCRUSH: AUTOMATIC BEAN POD EXTRACTING MACHINE

INTEGRATED DESIGN
PROJECT MCTA 3330

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GROUP 10

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Abstract

This report presents the design and development of an automated bean pod extracting machine for processing fresh edamame. The system is engineered within a compact structural envelope using a lightweight frame made of aluminum extrusions, acrylic shielding, and custom 3D-printed internal components.

The core extraction mechanism features two counter-rotating rollers covered in sandpaper to maximize grip and pull the pods through a fixed gap for cracking. The automation architecture integrates a low-speed geared DC motor for extraction, a high-speed conveyor belt controlled via PWM for transport, a vibrating motor for bean-hull separation, and a high-torque servo motor to regulate feeding at the hopper gate. For reliable operational safety, the entire mechatronic network is powered by an industrial AC-to-DC switching power supply integrated with a master emergency stop circuit. Analytical modeling and dynamic MATLAB simulations verify that this hardware configuration achieves automated hull separation safely, smoothly, and reliably.

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1.0 PROBLEM ANALYSIS

1.1 Problem Definition & Formulation

Traditional edamame shelling methods require a lot of manual labor and do not adapt to the different sizes of the pods. Because edamame pods are natural materials, they are highly unpredictable. For example, some are thick, some are small, some are soft, and others are tough. Without a way to monitor the machine's workload or changing pod sizes, operators cannot tell when the system is straining. This lack of tracking leads to major failures: tough pods cause sudden mechanical jams, motors overheat, and the soft edamame beans end up crushed and wasted. The impacts of these failures are severe, resulting in ruined crops, financial losses for farming families, and high food waste.

Podcrush: Automatic Edamame Pod Extracting Machine was developed as a fully functional, compact engineering solution specifically designed for small-scale operations and home-based businesses. Instead of relying on large, expensive industrial machinery, this system provides an affordable, space-saving appliance that combines a counter-rotating roller mechanism, an online tracking dashboard, and a remote Telegram control interface. It features built-in electrical safety monitoring to protect the propulsion drive, automatically tracking system strain to prevent motor burnout and alert the operator during heavy jams.

During the final project presentation, the examination panel validated the core functionality, cost-effectiveness, and usability of the design for small-scale users. To transition this successful prototype into a commercially viable product, three critical engineering pathways were highlighted for future optimization. First, the design must be evaluated to explore how the compact mechanism can be adjusted to process other types of beans or nuts. Second, because the current system is a development prototype, a strategy is required to upgrade the structural and contact components to certified food-safe grades. Finally, a standardized maintenance plan must be established to ensure long-term operational reliability in dusty and moisture-rich agricultural environments.

Consequently, the finalized engineering problem is defined as upgrading a functional, small-scale agricultural prototype into a versatile, commercial-grade food processing appliance. The goal is to evaluate the mechanical adaptability of the compact roller system for multi-crop processing, implement certified food-safe material alternatives, and define a clear maintenance framework, thereby satisfying strict food industry standards and long-term reliability requirements without compromising the machine's existing affordability, portability, safety monitoring, or remote control features.

1.2 literature review

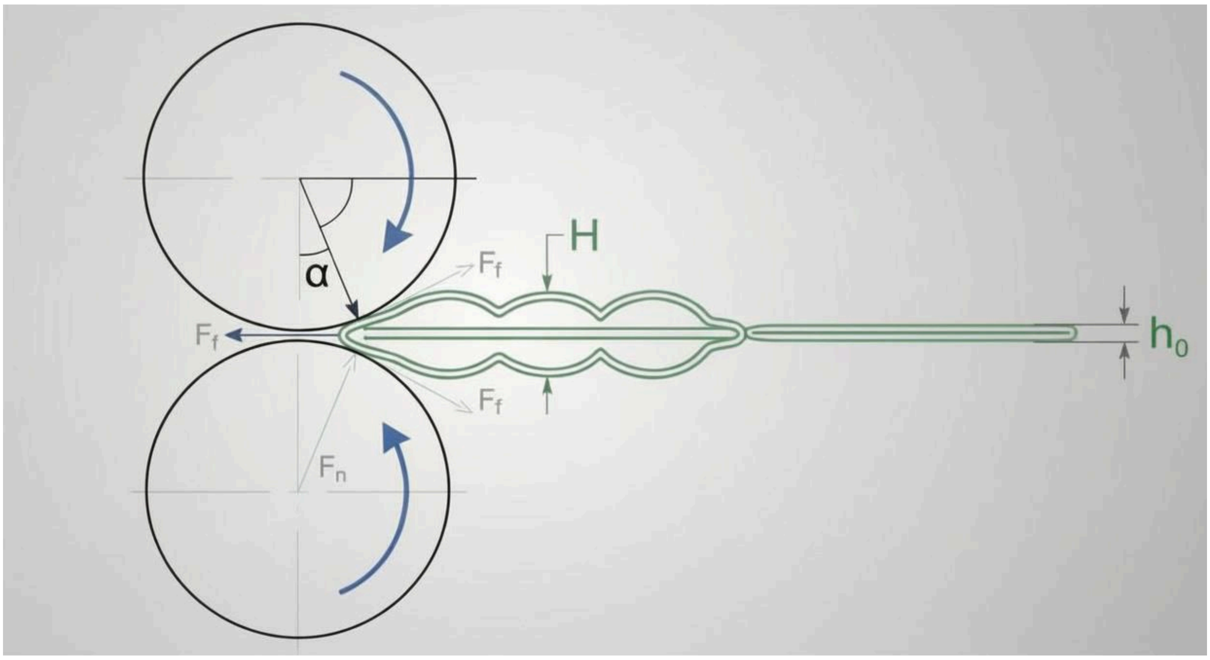
Getting the beans out of edamame pods requires a careful balance of squeezing and pulling forces. Because edamame is harvested while it is still green, the pods are soft and hold a lot of water, usually between 65% and 70%(Nair et al., 2023). Large industrial factories use massive, expensive drum machines to shell crops. While these work great for huge farms, they take up too much space, use a lot of power, and operate blindly without giving the user any live updates. For small-scale users and home businesses, there really hasn't been a smart, data-driven machine available on the market.

To validate this design space, our prototype was benchmarked against real-world market alternatives. For example, household electric strippers like the GT50X offer a compact, motorized table-top footprint, but they operate completely blindly using simple on/off switches with zero mechanical data feedback. On the mid-scale commercial side, systems like the Fengxiang DPL-300 implement full food-safe compliance and an adjustable shaft clearance system to handle varying pod profiles, but they remain heavy, rigid factory units with no remote monitoring capabilities. For true multi-crop processing, commercial machines like the Topp Machinery TPP-D1 utilize highly specialized adjustable disc spaces to peel multiple legume types (such as chickpeas, soybeans, and peas) without causing seed damage, showing that variable clearance control is essential for processing diverse agricultural inputs.

Our compact edamame extractor prototype bridges the gap between these systems. It fits right on a desktop for small-scale operations while combining a counter-rotating roller mechanism with an online tracking dashboard and a remote Telegram interface. Instead of using a complicated and expensive setup of physical distance sensors near the rollers, the machine uses integrated motor current sensing for safety. By tracking the motor's power draw, the system can automatically spot when the machine is working too hard, detect torque jumps, and stop the motor immediately during a heavy jam before the drivetrain burns out

To take this working prototype and upgrade it into a commercial food appliance as highlighted by our examination panel, our engineering trajectory focuses on two main standards. First, any parts touching the food must be upgraded to certified food-safe materials. Following commercial benchmarks like the DPL-300, this means replacing standard prototype structural framing with rust-resistant AISI 304 stainless steel and coating the rollers with food-grade Nitrile (NBR) rubber. This ensures the machine fully complies with Part VI (Packages and Appliances for Food) of the Malaysian Food Regulations 1985, which strictly prohibits the use of components that transfer harmful residues or compromise hygiene during contact with food (2024). Second, we can use the live current data sent to our IoT network to plan routine maintenance, making it easy to spot worn-out bearings or misaligned rollers before the machine breaks down.

1.3 Mathematical Modeling & Simulations



1.3.1 Initial System Constants & Assumptions

- **Roller Diameter (D):** 34 mm
- **Roller Radius (R):** $D/2$ = 17 mm = 0.017 m
- **Average Edamame Pod Thickness (H):** 10 mm = 0.010 m
- **Selected Minimum Gap Clearance (h_0):** 7.5 mm = 0.0075 m
- **Target Operational Roller Speed (N):** 9.6 RPM (Rated speed of the chosen motor)
- **Motor Rated Torque (T_motor):** 9 kgf.cm = 0.8826 N.m (from manufacturer datasheet)
- **Adjusted Coefficient of Friction (μ):** 0.75 (high-grit abrasive sandpaper covering)

1.3.2 Compression and Extraction Force Calculation

The geometric boundaries dictate the **nip angle (α)** at which the edamame pod initially makes contact with both rotating surfaces.

$$\cos\alpha = 1 - (H - h\_0) / 2R$$

$$cos\,\alpha = 1 - (10 - 7.5)/ 2(17) = 0.92647$$

$$\alpha = arccos\,(0.92647) = 22.11\,°$$

Verification of the Self-Feeding Gripping Condition:

For the machine to pull the edamame pod smoothly through the chamber without slipping, the friction condition must satisfy the equilibrium:

$\mu \geq \tan(\alpha)$

Evaluating both sides of the equation with the new abrasive material property:

$\tan(22.1°) = 0.406$

Condition Check: $0.75 \geq 0.406$ (Highly Satisfied)

1.3.3 Torque-Limited Normal Force

1. Maximum Allowable Normal Force (F_n):

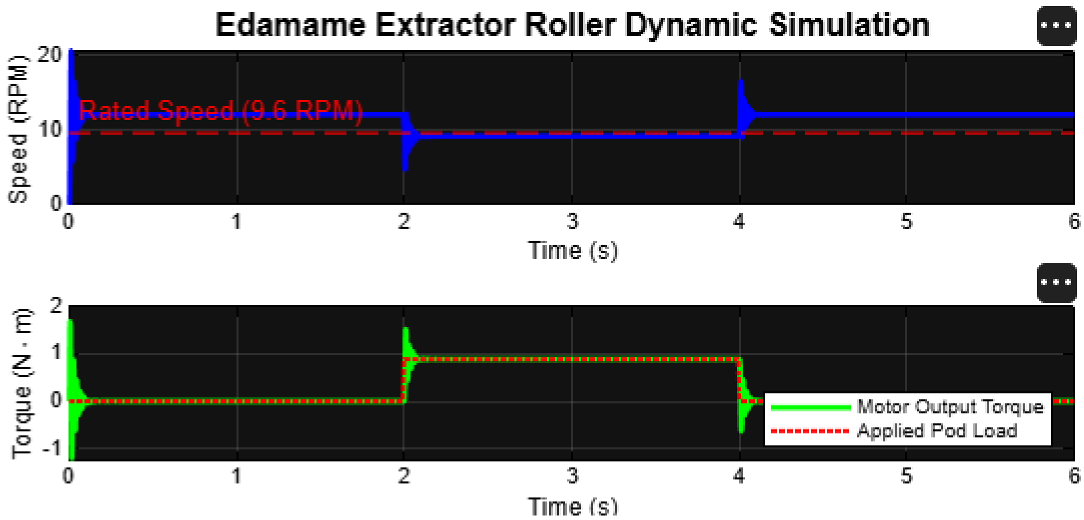
To ensure the system operates within safe electrical limits and prevents motor stalling, the datasheet units for torque (kgf.cm) are converted to SI units (N.m):

Using the torque relation and incorporating the agricultural equipment safety factor (S_f = 1.5):

$F_n = T_{motor} / 2(\mu)(R)(S_f)$

$F_n = 0.8826/2(0.75)(0.017)(1.5) = 23.07$ N (Maximum amount of force the extracting mechanism can provide)

System Simulation Analysis



To validate the operational stability of the selected 12V DC geared motor under non-continuous loading conditions, a state-space dynamic simulation was executed in MATLAB. The system behavior is characterized across three distinct operational phases:

No-Load Idle Phase (t = 0.0 s - 2.0 s): Following initial startup transients, the motor overcomes rotor inertia (J) and internal viscous damping (b) to achieve a stable steady-state no-load angular velocity of 12.0 RPM. The matching motor output torque remains at a negligible baseline level.

Pod Engagement & Extraction Phase (t = 2.0 s - 4.0 s): At t = 2.0 s, a step-load torque of 0.8826 N.m} (equivalent to 9.0 kgf.cm) is introduced, representing the physical resistance of an incoming edamame pod hull against the 34 mm sandpaper rollers. The sudden mechanical shock causes a brief deceleration transient before the motor's internal magnetic torque field balances the load, stabilizing perfectly at the target rated operational speed of **9.6 RPM**.

Post-Extraction Recovery Phase (t = 4.0 s - 6.0 s): The moment the pod clears the minimum 7.5 mm nip gap clearance, the external load resistance completely drops. The system exhibits a smooth asymptotic recovery curve, returning safely back to the 12.0 RPM idle state without manifesting excessive speed overshoot or prolonged settling time oscillations.

1.4 Failure Modes, Risks, and Constraints

To systematically identify, quantify, and mitigate potential design flaws, a formal Failure Mode and Effects Analysis (FMEA) was constructed alongside a Fault Tree Analysis (FTA) for the automated extraction system. This section defines the operational limitations, design constraints, and potential failure points of the machine, prioritizing risks based on the Risk Priority Number (RPN), calculated as RPN=SOD, where S is Severity, O is Occurrence, and D is Detection.

System / Item / Process Step	**Potential Failure Mode**	**Potential Failure Effects**	**Severity**	**Causes**	**Occurrence**	**Current Controls**	**Detection**	**RPN**	**Recommended Actions**
Roller Friction / Traction	Surface slipage	Pods pass through whole; zero bean extraction	8	High moisture/ sap on pods reducing NBR friction	7	Visual inspection of discharge chute	2	112	Implement diamond-tread texture on NBR rollers
Gap adjustment (2-10 mm)	Gap expansion (creep)	Inconsistent splitting; efficiency drop	7	Vibration at high rpm (120) loosening manual screws	8	Hourly manual gap measurement cal	4	224	Install nylon-insert lock nuts or detent system
Roller core(AISI 304)	Shaft deflection	Crushed beans; uneven pressure across roller	9	Core diameter too small for high-volume loads	3	Post-extraction bean quality audit	5	135	Increase AISI 304 shaft gauge or add centre support
Motor drive/VFD	Mechanical jam	Gear shearing or motor winding burnout	10	Pod overcrowding or foreign debris in hopper	4	Manual emergency stop button	2	80	Program PLC current-sensing auto-reverse

Bearing housing	Bearing seizure	Total system shutdown; high repair cost	8	Plant sap and dust ingress into housing	5	Regular lubrication schedule	6	240	Upgrade to IP65-rated sealed stainless bearings

The 5 Whys method was applied to the two failure modes with the highest Risk Priority Numbers (RPN) to ensure the root causes rather than just the symptoms are addressed in the final machine design.

Analysis A: Premature Bearing Seizure (RPN 240)

- **Problem:** The extraction rollers stop rotating, and the motor draws excessive current.
- **Why 1?** The bearing internal components have seized due to corrosion and debris.
- **Why 2?** Plant sap and fine dust from the bean pods entered the bearing housing.
- **Why 3?** The current bearing units are not adequately sealed against liquid or fine particulate ingress.
- **Why 4?** The design used standard shielded bearings instead of environmentally sealed units.
- **Why 5 (Root Cause):** The environmental sealing specification did not account for the organic "slurry" (moisture + sap + dust) inherent in agricultural processing.
- **Countermeasure:** Upgrade to IP65-rated sealed stainless steel bearings to prevent contamination ingress.

Analysis B: Inconsistent Extraction (Gap Expansion - RPN 224)

- **Problem:** The roller gap drifts beyond the 10mm limit, letting pods pass through whole.
- **Why 1?** The manual adjustment screws rotate during operation.
- **Why 2?** The assembly is subject to high-frequency resonance at speeds of 120 RPM.
- **Why 3?** The standard screw threads lack sufficient friction to resist vibration-induced torque.
- **Why 4?** The design relies solely on the operator's manual tightening to hold the gap position.
- **Why 5 (Root Cause):** The adjustment mechanism lacks a positive-locking feature to maintain the 2mm–10mm geometry under vibration.
- **Countermeasure:** Implement a spring-loaded detent system or Nylon-insert lock nuts to secure the adjustment shaft.

System Constraints

While the FMEA and FTA methodologies address the operational risks of the extraction mechanism, the overall design and development of the machine are strictly governed by specific project limitations. As an academic proof-of-concept, the system architecture must balance theoretical reliability with practical fabrication boundaries. The following system constraints define the conditions under which this prototype was engineered:

- **Academic Budget Limitations:** As a student-funded, proof-of-concept prototype, component selection is strictly constrained by budget. This necessitates the use of commercially available, off-the-shelf components such as a standard 24V geared DC motor, BTS 7960 motor driver, and an ESP32 microcontroller rather than high-cost, industrial-grade PLCs or servomotors.

- **Fabrication and Machining Constraints:** Manufacturing is limited to the accessible tools within the university workshop. Consequently, the design is constrained to utilize standardized modular parts, such as 40mm x 40mm aluminum extrusions for the main frame and 3D-printed sub-components, to eliminate the need for expensive custom CNC machining.

- **Control Logic Constraints:** The integration of the ESP32 dictates that all control logic operates at 3.3V. This requires careful voltage level considerations when interfacing with the 24V industrial sensors and motor driver circuitry to prevent logic-level degradation or microcontroller damage.

- **Torque & Power Constraints:** Operating within the limitations of the selected off-the-shelf hardware, the 24V DC high-torque geared motor must operate at an average load of 20-30% of its stall torque to prevent thermal runaway during the testing phases.

- **Environmental Constraints:** Despite being a prototype, the machine must still handle organic "slurry" (moisture, plant sap, and soil) during physical testing, necessitating a minimum Ingress Protection approach, prioritizing IP65-rated mechanical bearings where possible.

2.0 Design Documentation & Justification

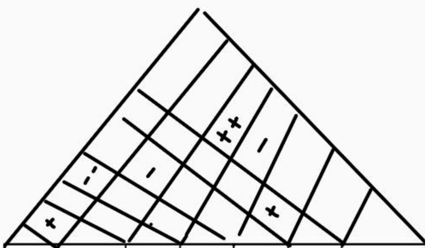
2.1 Conceptual Design & Selection Methodology

To ensure that the development of the automated bean pod extraction machine aligns closely with customer expectations while remaining mechanically robust, a structured, two-staged design selection framework was implemented.

First, a Quality Function Deployment (QFD) framework was utilized to translate qualitative customer requirements into quantifiable engineering characteristics, establishing design priorities. Second, a Pugh Concept Screening Matrix was deployed to objectively evaluate alternative design variants against a baseline configuration (Datum), highlighting the most viable architectural choice for subsequent development.

Quality Function Deployment (QFD) & House of Quality

The House of Quality (HoQ) matrix correlates user expectations with engineering metrics using standard relationship multipliers: Strong (9), Moderate (3), Weak (1), or None (0).



Customer Requirements \ Design Requirements	Importance	motor speed (RPM)	extraction mechanism design	roller material	power consumption	safety guard & enclosure	control system & user interface	machine structure and frame
fast pod extraction	5	9	9	3	3		1	
high extraction efficiency	5		9	9	3			
easy to operate	4	1	3				9	3
safe to use	5		1			9	3	3
low maintanance	4	1	3	9		1	1	3
durabilty	4			9				9
reliabality	5	1	3	9		1	1	9
raw score		58	134	177	30	54	65	120
relative weight		9%	20%	27%	5%	8%	10%	18%
rank order		5	2	1	7	6	4	3

By quantifying the relationship between customer expectations and engineering metrics, the QFD matrix effectively isolates the core technical drivers of the bean pod extraction machine. The resulting relative weights mathematically highlight where design optimization will yield the greatest impact on machine performance, longevity, and operational safety. A critical analysis of the top-ranking engineering characteristics reveals two primary technical mandates for the system's architecture:

- Material and Mechanical Dominance: Blade/Roller Material (26.6%) and Extraction Mechanism Design (20.1%) remain the primary engineering priorities. This underscores that achieving excellent extraction efficiency while minimizing bean damage depends fundamentally on the physical contact dynamics of the rollers.
- Structural Safety Factor: Isolating Reliability and Durability caused a significant math escalation in the importance of the Machine Structure & Frame (rising to 18.0%). This mathematically justifies the engineering need for a rigid, vibration-resistant frame structure to ensure the machine stands up to rigorous, long-term agricultural usage without structural fatigue.

<u>Pugh Concept Screening and Selection</u>

Following the identification of the engineering priorities, four distinct design variants (Concepts 1, 2, 3, and 5) were evaluated. Concept 3 was selected as the Datum (baseline) against which all other options were evaluated using weighted performance criteria.

The screening criteria weights reflect the engineering weights derived from the QFD phase, focusing heavily on quality of extraction (10), throughput rate (9), and extraction efficiency (8).

		1	2	3	4
		concept			
Criteria	**weight**	3	1	2	5
extraction efficiency	8		+	-	+
probability of jamming	6		+	-	-
durability	5		-	+	+
cost of materials	5		+	+	-
weight	3		-	+	-
safety	6	**datum**	+	-	+
maintanance time	4		-	-	-
throughput rate	9		-	+	+
storage volume	2		+	+	-
quality of extraction	10		+	+	+
totals of pluses			6	6	5
Totals of minuses			4	4	5
weighted score			16	10	18

Pugh chart

- Concept 3 (The Datum): A standard, single-speed, direct-drive extraction system utilizing a basic rigid gravity chute. It represents a middle-of-the-road market baseline constructed from un-coated Carbon Steel and Mild Steel.
- Concept 1 (Lightweight Cost-Optimized Variant): A lightweight configuration utilizing a single motorized roller pressing against a stationary plate. It relied on 6061-T6 Aluminum extrusions for the frame and Polycarbonate/PVC for the scraping elements to minimize initial costs.
- Concept 2 (High-Durability Bulk Variant): A heavy-duty configuration built from thick Galvanized Plate Steel and Cast Iron crushing blocks. It featured a massive storage hopper designed for brute-force bulk loading but suffered from a heavy structural footprint and high mechanical resistance.
- Concept 5 (Single-Motor High-Torque System): A streamlined mechatronic design utilizing a single high-torque DC gear motor to actively drive a primary roller, while the secondary roller acts as a passive friction-driven idler. Optimized for rapid prototyping, its structure integrates an aluminum frame, 3D-printed (PLA) internal mechanics, and transparent acrylic safety guards.

Based on the total calculated weighted scores, Concept 5 achieved the highest ranking with a score of 18, making it the mathematically and operationally superior choice for the final physical machine architecture.

While Concept 1 offered advantages in entry-level material costing and lower initial weight, and Concept 2 provided unparalleled bulk storage, Concept 5 heavily outperformed the alternatives within the most critical performance criteria: Throughput Rate (+), Extraction Efficiency (+), and Quality of Extraction (+).

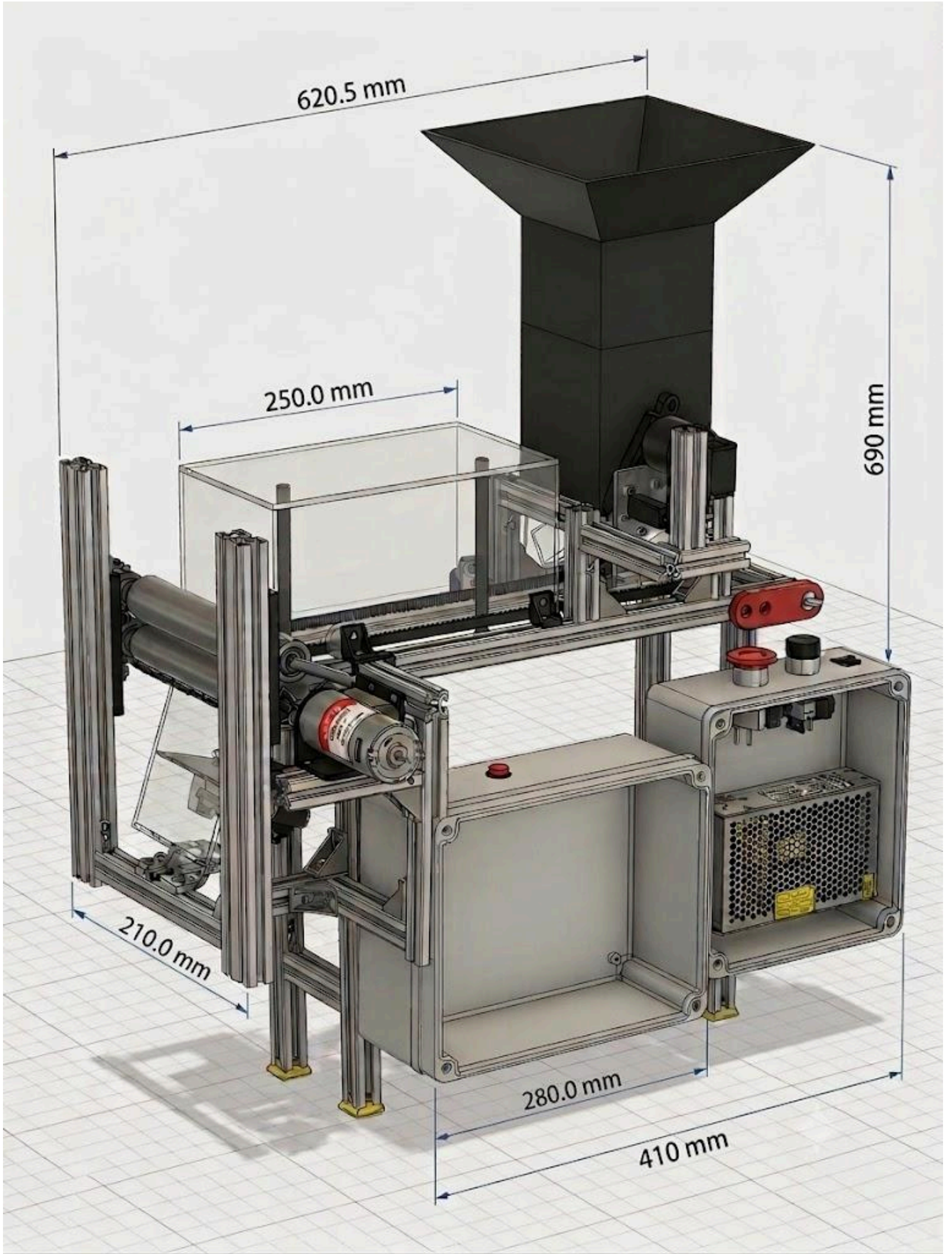
The compromises of a higher probability of jamming (-) and increased structural complexity due to the single-motor high-torque configuration were successfully mitigated during the prototype design phase through two key engineering strategies:

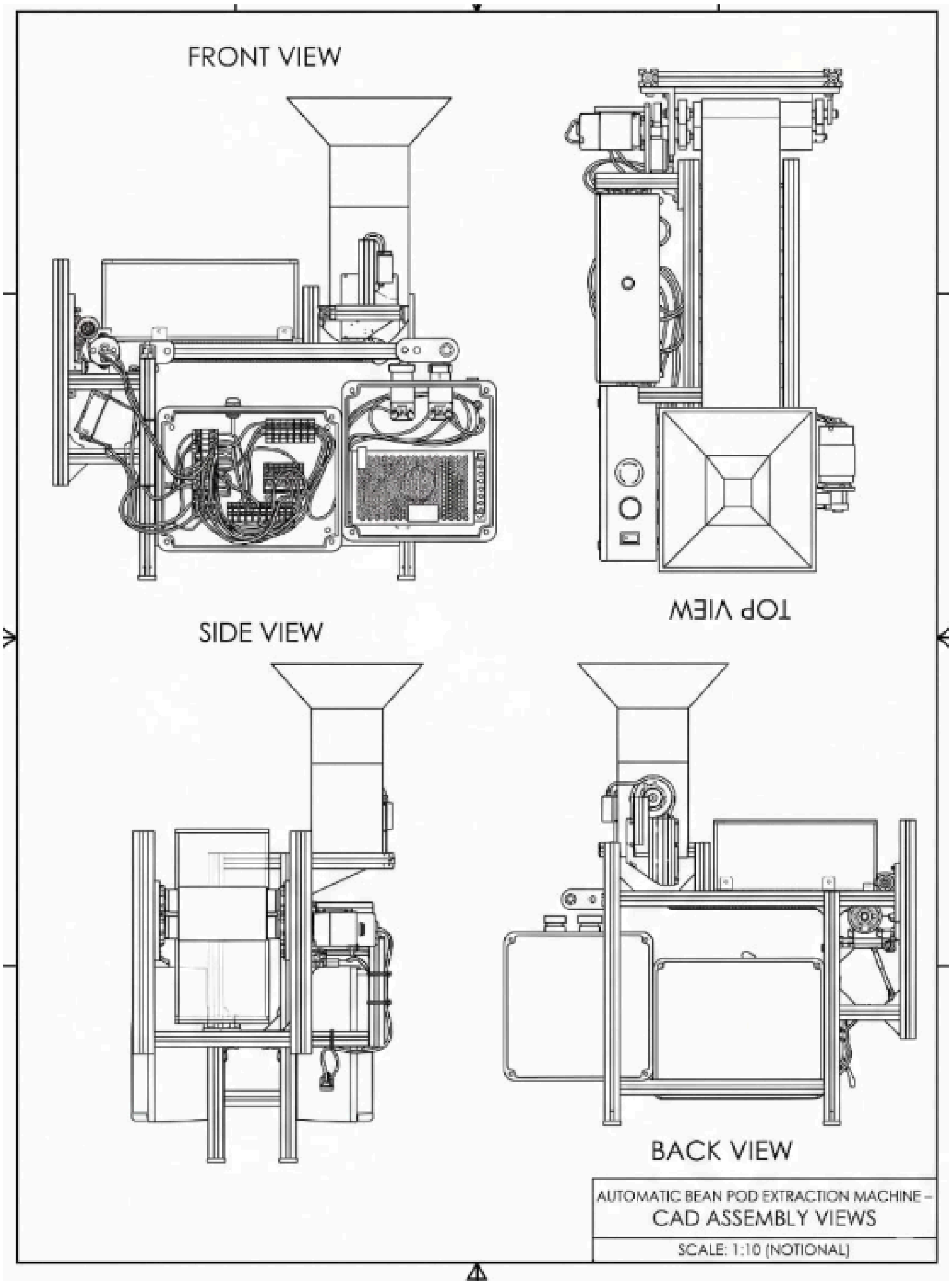
- Visual Monitoring and Safety: By utilizing transparent Acrylic enclosures, any operational feedstock jams can be visually identified and cleared safely by the operator, neutralizing the operational risk of the single-driven roller configuration.
- Design for Serviceability (DFS): To offset maintenance concerns, the system was engineered almost entirely around standardized, commercial off-the-shelf (COTS) components. Rather than relying on custom-machined proprietary hardware, the assembly utilizes standard metric fasteners (screws, nuts) and a universally available, standard-framed high-torque DC gear motor. This ensures that if any wear or unexpected hardware failure occurs during operation, components can be sourced locally and replaced immediately, minimizing operational downtime.

Ultimately, Concept 5 delivers the ideal balance between high-efficiency agricultural processing and a modular, easily maintainable mechatronic architecture using an Aluminum frame and 3D-printed PLA internals, making it uniquely suited for the final physical build.

2.2 Design Documentation & Architecture

2.21 3D-CAD Models




FRONT VIEW
TOP VIEW
SIDE VIEW
BACK VIEW
AUTOMATIC BEAN POD EXTRACTION MACHINE –
CAD ASSEMBLY VIEWS
SCALE: 1:10 (NOTIONAL)

2.22 System Architecture Diagrams

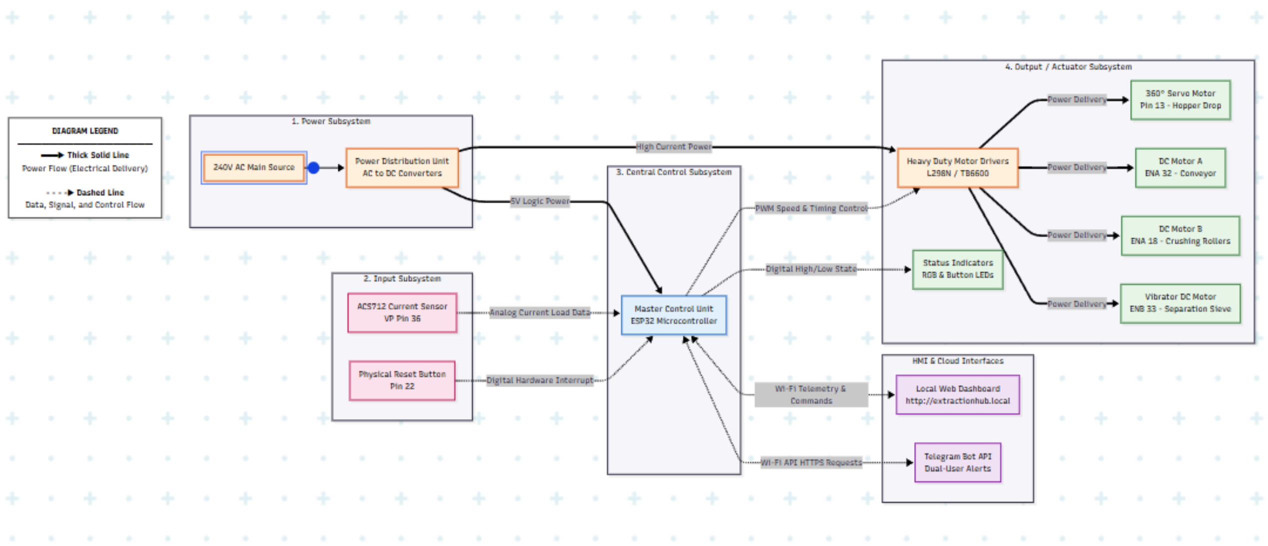


Figure 2.1

The system architecture of the Automatic Bean Pod Extraction Machine integrates mechanical actuators, electrical power distribution, and intelligent software control into a unified mechatronic framework. As illustrated in the system architecture diagram, the design is divided into four primary subsystems, alongside a cloud-based network interface.

The Fig. 2.1 explicitly differentiates between **Power Flow** (thick solid lines), representing high-current electrical energy delivery, and **Data/Signal Flow** (dashed lines), representing low-voltage communication, PWM, and sensor data.

1. Power Subsystem The system is powered by a standard 240V AC main source. This power is routed through a Power Distribution Unit (PDU) containing AC-to-DC converters. The PDU steps down the voltage and splits it into isolated power buses: a stable 5V bus to safely power the logic of the microcontroller, and higher-current buses (e.g., 12V/24V) routed directly to the motor drivers to handle the heavy mechanical loads of the extraction process.

2. Input Subsystem This subsystem acts as the sensory network for the machine.

- **ACS712 Current Sensor (Pin 36):** Continuously monitors the electrical load drawn by the extraction motors. If the load spikes above the 2.9A threshold, it signals a mechanical jam to the MCU.
- **Physical Reset Button (Pin 22):** Provides the operator with a localized, hardware-level interrupt to manually clear faults and reset the machine's operational state.

3. Central Control Subsystem (The Brain) The core of the system is the **ESP32 Microcontroller**. It acts as the master processing unit, simultaneously handling hardware-level I/O tasks and network communication. The ESP32 processes analog sensor data, applies control logic, and executes the strict hardware timing required (such as the 360-degree servo locking logic) to prevent material overflow.

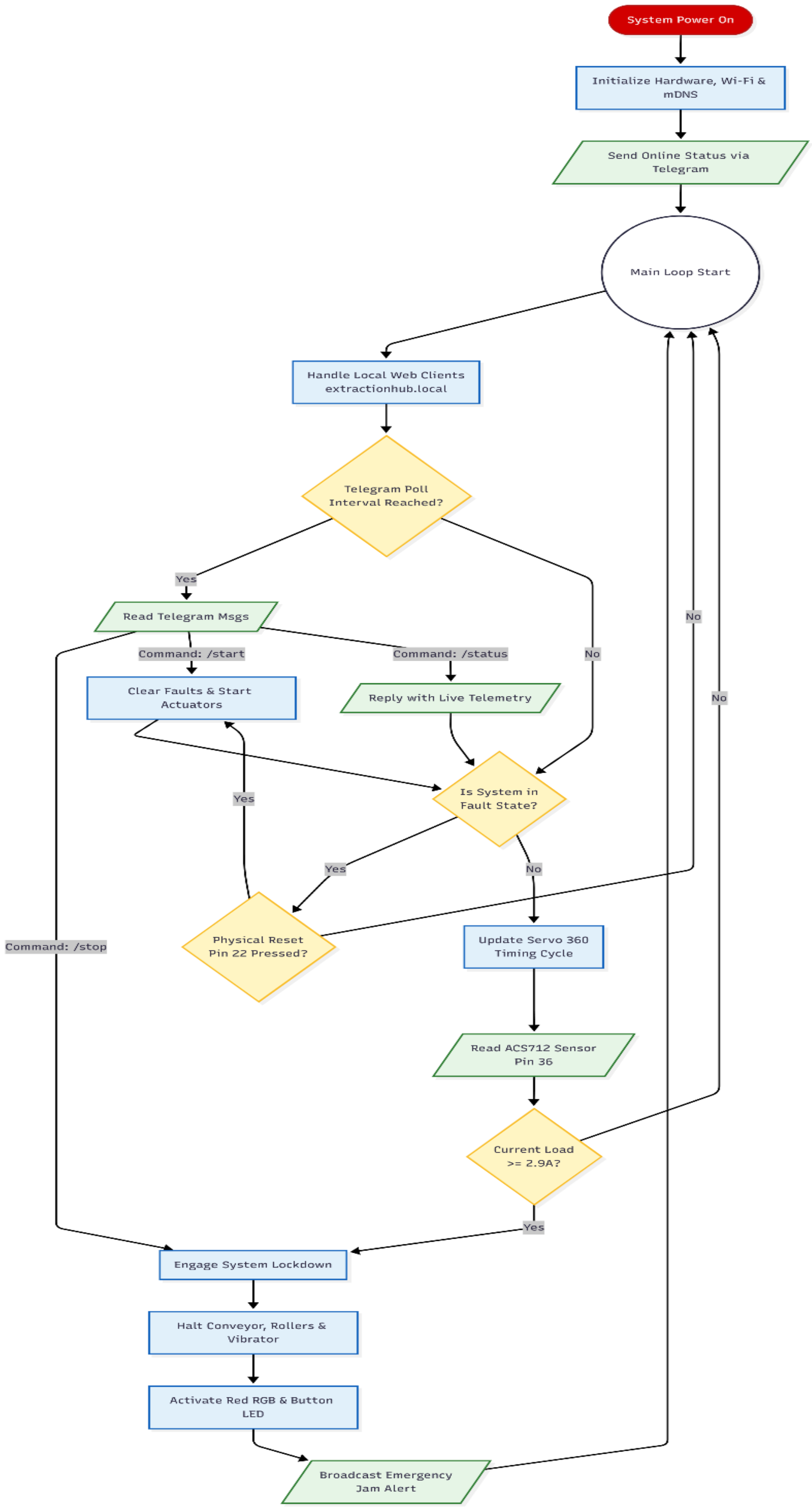
4. Output / Actuator Subsystem The ESP32 sends low-voltage Pulse Width Modulation (PWM) and digital signals to heavy-duty Motor Drivers (e.g., L298N, TB6600). These drivers act as heavily controlled electrical gates, delivering high-current power to the physical actuators:

- **360° Servo Motor (Pin 13):** Controls the rotational drop rate of raw pods from the hopper.
- **DC Motor A (Pin 32):** Drives the transport conveyor belt at maximum speed.
- **DC Motor B (Pin 18):** Powers the high-torque crushing rollers for the actual extraction.
- **Vibrator DC Motor (Pin 33):** Shakes the separation sieve to isolate the extracted beans from the waste pods.
- **Status Indicators:** RGB and Button LEDs provide immediate, localized visual feedback on machine health (e.g., green for active, red for lockdown).

5. HMI & Cloud Interfaces Because the ESP32 features built-in Wi-Fi, the machine bypasses traditional wired control panels. It broadcasts live load telemetry and machine state data to two distinct endpoints:

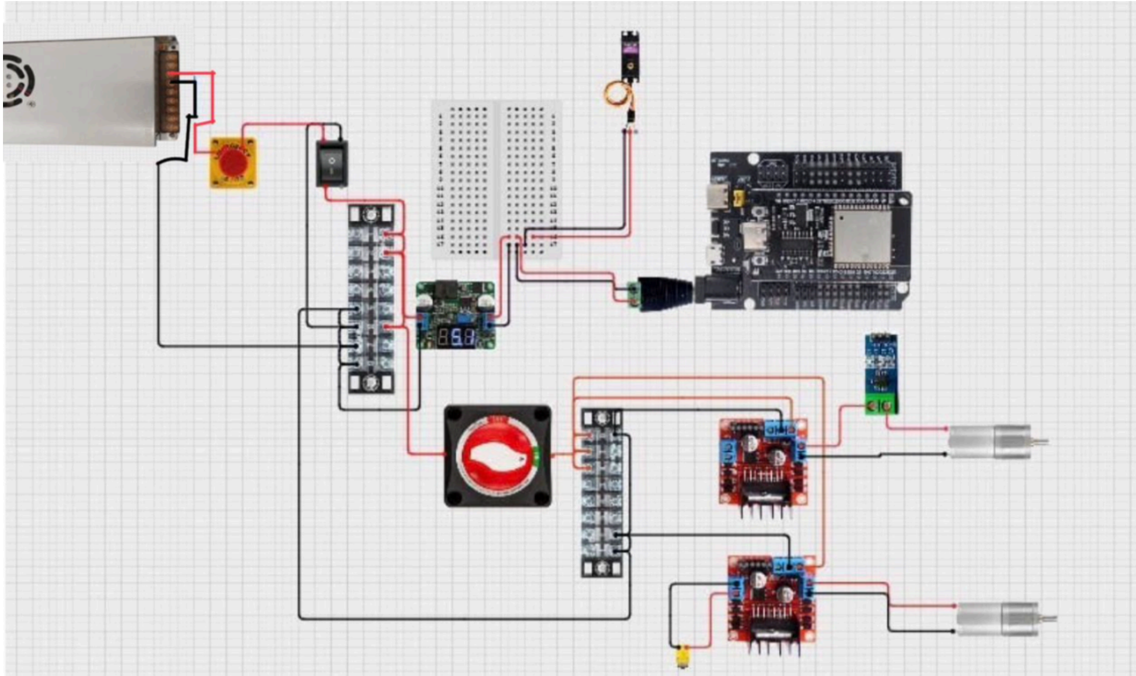
- **Local Web Dashboard:** Accessible via http://extractionhub.local, allowing operators on the same network to remotely start, stop, and monitor the machine.
- **Telegram Bot API:** Sends automated, dual-user push notifications to mobile devices in the event of an emergency (e.g., a current spike lockdown) and accepts remote override commands.

2.23 Flowcharts

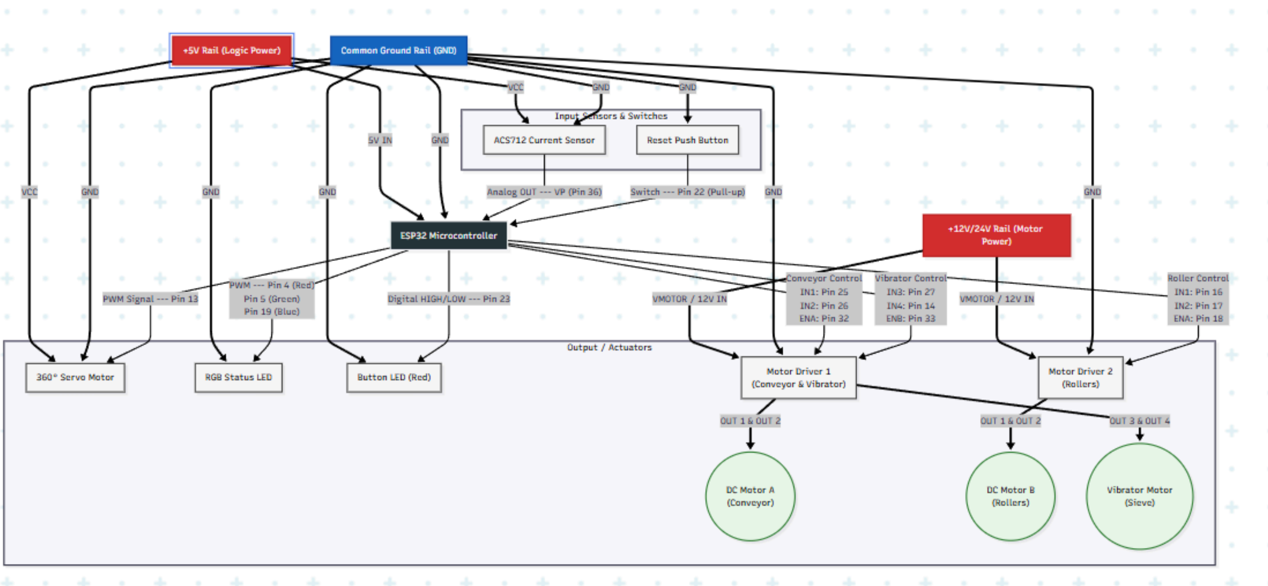


2.3 Engineering Schematics

2.31 Circuit Diagrams



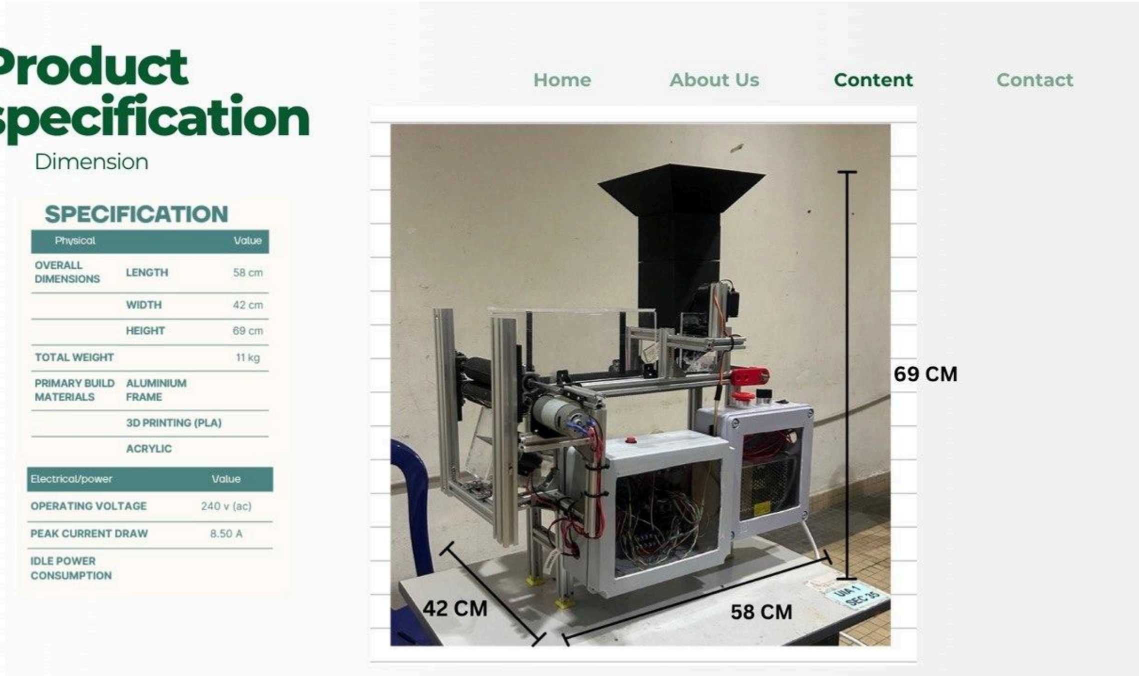
2.32 Design Schematics



2.4 Component Selection & Sizing Calculations

To realize the physical design of the Automatic Bean Pod Extracting Machine, the component selections must be justified through structural, volumetric, and electrical calculations that satisfy the strict boundary constraints of the project brief.

2.4.1 Volumetric & Geometric Sizing



The structural frame must house all components, processing zones, and electrical tracks while remaining strictly within the maximum allowed structural envelope:

- Maximum Length = 580 mm = 0.58 m
- Maximum Width = 420 mm = 0.42 m
- Maximum Height = 690 mm = 0.69 m
- Mass = 11.0 kg

2.4.2 Frame and Structural Material Selection Rationale

A multi-material hybrid design framework is selected to optimize the strength-to-weight ratio, ensuring total mass does not exceed 11 kg

1. Main Structural Skeleton (Aluminum 2020 T-Slot Extrusions): Selected for the primary load-bearing chassis. Aluminum (2.7 g/ cm^3) provides high structural rigidity to resist the 23.07 N separating forces generated between the rollers while maintaining a low weight profile compared to mild steel (7.85 g/ cm^3).
2. Machine Enclosure Panels (Clear Acrylic Polycarbonate): Selected for safety shielding and aesthetic visibility. It allows real-time visual inspection of the extraction zone and protects operators from flying plant debris while meeting food-safety standards.

3. Feeding Mechanism, filter and Gears (Fused Deposition Modeling - 3D Printed PLA): High-density Polyethylene/PLA is chosen for complex geometries like the bean collecting filter and load-bearing gear to reduce custom machining costs and minimize mass.

2.4.3 Actuator Selections

1. Extraction motor: 12V DC Low-Speed Geared Motor

- **Mechanical Function:** Drives the main 34 mm counter-rotating rollers to push the edamame bean out of its pod.
- **Selection Justification:** Based on the structural models in Section 1.3, high torque at low speeds is required to prevent slipping and manage the abrasive coefficient of friction ($\boldsymbol{\mu} = 0.75$). The 12 RPM No-Load (9.6 RPM Rated) model is selected. It outputs a high rated torque of 0.8826 N.m (9.0 kgf.cm) and an ultimate peak stall torque of 1.471 N.m (15.0 kgf.cm). This provides up to 57.69 N of transient pinching force, which safely exceeds the documented pod-cracking threshold of 13.4 N - 20 N while running at a stable 0.35 A current profile.

2. Transport Assembly: 12V DC High-Torque Geared Motor (Conveyor)

- **Mechanical Function:** Operates the drive pulley of the intake/outtake conveyor belt to transport pods smoothly between feeding mechanism and extracting mechanism.
- **Selection Choice:** A 12V DC Geared Motor rated at 500 RPM (Under Load) with a rated torque output of 0.5 kgf.cm**.**

. Separating Mechanism: 12V DC Eccentric Rotating Mass (ERM) Vibrating Motor

- **Mechanical Function:** Agitates the slanted separating plate to isolate the extracted beans from the shredded hull debris via gravity and size exclusion.
- **Selection Justification:** Legume hulls tend to stick to surfaces due to internal moisture and sap. To break these adhesive surface tension forces, a high-frequency, low-amplitude vibration is required. A 12V DC ERM Vibrating Motor (operating at 6000 RPM) is mounted directly beneath the 3D-printed sorting tray. The high rotational speed of the unbalanced internal mass creates enough continuous multi-axis acceleration to fluidize the bean-and-chute mixture, causing the dense beans to slide into the collection bin while hulls are guided out the waste chute.

4. Feeding Control / Stopper Mechanism: High-Torque Digital Servo Motor (Mg996R)

- **Mechanical Function:** Acts as an automated gatekeeper/gate indexer at the hopper base, rotating a 3D-printed flipper or pin to feed pods into the conveyor line one by one.
- **Selection Justification:** Unlike the continuous-rotation motors, the feeding gate requires exact angular positioning (0° closed, 45° open) and high holding torque to resist the weight of piled pods in the hopper. A High-Torque Digital Servo (MG996R) is selected operating at a standard 5 VDC logic supply. The torque capacity prevents

bean pods from forcing the gate open, while its internal closed-loop potentiometer allows the system esp32 to control precise feed intervals.

2.4.4 Component Selections

To ensure structural and electrical reliability under peak crushing loads, components were selected based on their deterministic performance parameters, thermal thresholds, and communication bandwidth.

- **Processing Core (ESP32 DevKitC):** An 8-bit architecture (e.g., ATmega328P) was rejected due to thread-blocking limitations. The ESP32's **Dual-Core 32-bit Tensilica LX6 architecture** (240 MHz) provides the processing bandwidth necessary to manage concurrent operations. Core 0 is dedicated to high-latency network overhead (secure TLS/SSL cloud handshakes for the Telegram Bot API), while Core 1 handles real-time deterministic tasks (high-frequency PWM generation, analog sensor sampling, and hardware interrupts). This parallel distribution ensures that secure network communications never introduce execution lag into critical physical safety loops.
- **Current Sensing Module (ACS712-20A):** Selected due to its integrated Hall-effect transducer path, which features an exceptionally low internal conductor resistance of 1.2 m Omega. This minimizes I^2R ohmic power loss and prevents localized thermal build-up inside the sealed control enclosure. Crucially, it provides up to 2.1 kV RMS galvanic isolation, shielding the low-voltage 3.3V ESP32 logic pins from the high-voltage inductive back-EMF spikes generated by the motor armatures.
- **Actuator Power Drivers:** An integrated dual H-bridge motor driver (L298N) configuration was deployed to isolate the electrical noise profiles of the high-current conveyor line, the vibration unit, and the extraction rollers. The driver gates are rated to survive the thermal profiles of sustained high-torque crushing loads.

2.4.5 Software Selections

The firmware for the "PodCrush Automatic" system is engineered using a hybrid time-slicing architecture. To manage dual communication layers alongside physical hardware, the microcontroller utilizes an execution loop regulated by millisecond clock differentials (millis). This distributes processing power across local network management, cloud polling, and safety telemetry without causing total system stalls.

1. Dual-Layer Communication Architecture:

The software operates two distinct human-machine interface (HMI) layers concurrently:

- **Edge-Localized Web Server:** The ESP32 hosts a native HTTP server utilizing Multicast DNS (mDNS). By broadcasting the domain https://extractionhub.local, it provides a low-latency control dashboard for on-site operators. The dashboard executes a dynamic meta-refresh every 2 seconds to fetch live telemetry without requiring manual browser reloads.

- **Secure Cloud Telemetry (Telegram API):** The firmware utilizes the UniversalTelegramBot library wrapped in a secure SSL client. To prevent continuous cloud polling from overloading the processor and lagging the local web server, the software implements a time-sliced execution window (telegramInterval). The microcontroller restricts Telegram server queries to exactly once every 1000ms.

2. Hybrid Execution Loop & Multitasking Strategy:

Because standard microcontrollers execute code sequentially, the firmware divides the main loop() into prioritized task blocks evaluated against internal timers:

- **Priority 1 - Web Client Servicing: server.handleClient()** is executed continuously at the top of the loop to ensure high responsiveness to local operator dashboard clicks.

- **Priority 2 - Actuator Escapement:** The gating servo mechanism is controlled using a hybrid timing state machine (updateServo360TimeBased()). It evaluates a background clock to trigger the open/close phases (800ms dwell open, 5000ms dwell closed). During the actual physical actuation, a strict 180ms hardware lock (delay(180)) is utilized. This ensures the servo completes its rotational stroke perfectly before yielding processing power back to the network stack, preventing network interruptions from causing partial mechanical gate openings.

- **Priority 3 - Safety Polling:** The ACS712 current sensor is polled every 200ms. To filter out analog noise and inductive ripple from the DC motors, the firmware captures 20 rapid ADC samples per cycle, averages the raw digital values, and maps the result to an absolute Amperage value.

3. Fault Isolation & Emergency Execution Protocol:

The software utilizes the calculated 2.90A hardware limit to enforce strict digital safety parameters, preventing thermal runaway and power supply brownouts.

- **Inrush Current Bypass:** DC motors require massive starting currents to overcome static inertia. The firmware implements a 4500ms bypass window (SENSOR_BYPASS_TIME) upon boot. During this window, transient current spikes are monitored but intentionally ignored to prevent a false-positive system lockout.

- **Catastrophic Stall Mitigation:** Post-startup, if a sampled reading registers ≥ 2.9A, the stopSystem() routine triggers instantaneously. The software drops all PWM drive channels to zero (ledcWrite = 0), mechanically isolating the conveyor, vibrator, and extraction rollers.

- **Synchronous Alert Broadcasting:** Upon fault detection, the RGB status lamp shifts to red and the physical panel LED is latched high. The firmware immediately executes a secure broadcast, pushing the localized fault data (including the exact trip current and diagnostic origin) to two remote supervisory Telegram accounts simultaneously.

3.0 Impact Analysis & Standard

3.1 Societal, Health, and Safety Impacts

The deployment of the PodCrush Automatic machine introduces measurable impacts on occupational health, workplace safety, and broader agricultural processing workflows.

1. **Health and Safety Impacts:**
    a. The primary mechanical hazard is the in-running nip point generated by the counter-rotating extraction rollers, which creates a crushing hazard.
    b. Electrically, the integration of a 24V high-torque motor and BTS 7960 motor driver presents a thermal and electrical hazard if the system reaches near-stall conditions, which can lead to winding burnout or melted wiring terminations.
    c. The continuous oscillation of the vibrating separation platform and the 50Hz operational motor vibrations introduce acoustic noise and present an ergonomic hazard if vibrations transfer unchecked to the operator's workbench.
2. **Societal Impacts:**
    a. Automating the extraction process significantly reduces the ergonomic strain and repetitive motion injuries traditionally associated with manual pod splitting.
    b. By utilizing a continuous feed-and-monitor system, small-to-medium agricultural enterprises can increase their throughput while protecting workers from the physical toll of manual extraction.

3.2 Environmental Sustainability

To align with modern sustainable engineering practices, the machine's design incorporates principles of resource efficiency and extended product lifecycle management.

1. **Material Selection and Circularity:**
    a. The system heavily utilizes Design for Assembly (DFA) and Design for Manufacture (DFM) principles, opting for standardized fasteners instead of permanent industrial adhesives.
    b. This design for recycling ensures the machine can be easily disassembled at the end of its lifecycle.
    c. The primary roller shafts are constructed from recyclable AISI 304 Stainless Steel, and plastic is utilized for the waste collection bin to reduce fabrication costs.

2. **Waste Management and Mechanical Separation:**
    a. The machine intrinsically separates the final product from the organic waste via the vibrating separation platform.

b. The floor of the platform is filled with designated holes that allow only the extracted beans to pass through into the finished product collection.
c. The unwanted husks are directed into a dedicated waste bin, allowing the organic material to be easily collected and repurposed as residual biomass.

3. **Component Lifespan Optimization:**
    a. To prevent the machine from becoming premature electronic waste, specific environmental protections are engineered into the components.
    b. Using a silicone-specific primer (e.g., Dow Corning 1200 OS) prevents organic acids from delaminating the rubber rollers, while slinger rings protect the bearings from corrosive sap, vastly extending the operational lifespan of the hardware.

3.3 Safety Protocols & Mitigation Strategies

Rigorous mitigation strategies have been engineered into the system to neutralize the risks identified in the safety analysis and the root cause assessments.

1. **Hardware and Mechanical Safeguards:**
    a. **Mechanical Guarding:** The extraction rollers are deeply recessed within the main 40mm x 40mm aluminum extrusion frame and shielded by transparent acrylic panels. This ensures operator hands cannot reach the moving parts during operation while maintaining visual monitoring.
    b. **Emergency Stop:** A hardwired, twist-to-release Emergency Stop (E-Stop) button is prominently mounted on the main control enclosure to immediately cut power in the event of a mechanical jam or accidental contact.
    c. **Vibration Mitigation:** To counteract the continuous oscillation from the separating platform and the 50Hz motor vibrations, all critical adjustment points utilize dual-locking systems comprising DIN 985 Nylon-insert lock nuts and Loctite Blue 243 thread-locking fluid.
    d. **Stability Control:** The machine chassis is engineered with a low center of gravity and is supported by high-friction rubber feet to prevent it from shifting, moving, or falling from the workbench due to the internal vibrating separation mechanism.
2. **Active Current Monitoring & Anti-Jam Logic (Software Safeguard):**
    a. The system actively monitors the torque load on the 12VDC geared motor by reading the analog current draw through the motor driver.
    b. If a foreign object enters the extraction rollers, the motor attempts to draw stall current. The ESP32 is programmed with a maximum safe current threshold. If the high-resolution analog-to-digital converter (ADC) detects a current spike exceeding this limit, the logic triggers an immediate "Motor Halt" and executes a brief reverse sequence to eject the blockage, protecting the gear teeth from shearing.
3. **Dual-Core Processing & Watchdog Timer (Software Safeguard):**

   a. Leveraging the ESP32's dual-core architecture, the critical safety monitoring (like the E-Stop interrupt and current sensing) can be isolated to run concurrently with the main motor control loop.
   b. Additionally, a Hardware Watchdog Timer (WDT) is enabled. If the microcontroller hangs due to a power fluctuation or logic error, the WDT automatically forces a hard reset, returning all motor driver pins to a safe, de-energized (LOW) state.
4. **Visual Fault Indication (Operator Safeguard):**
   a. Given the acoustic noise of the vibrating separation platform, the control panel is equipped with high-visibility LED status indicators. A normal running state is indicated, but if the E-Stop is pressed, the current limit is tripped, or the system enters a safe-state default, a solid red fault indicator illuminates, requiring operator intervention.

3.4 Standards Compliance & Legal/IP Implications

To ensure the automated bean pod extraction machine meets global benchmarks for safety, interoperability, and operational quality, the design framework strictly adheres to specific international engineering standards across its mechanical, electrical, and materials systems

1.Mechanical Safety & Machinery Standards

- **ISO 12100:2010 (Safety of Machinery — General Principles for Design)**: This standard guided the risk assessment and hazard mitigation strategy for the machine. To isolate the high-torque extraction mechanism from human contact, the design utilizes an automated bulk feeding architecture. Operators load feedstock into an isolated hopper assembly, entirely removing the need for manual feeding near the active roller nip-point. Physical safeguarding is maintained via a transparent Acrylic safety guard enclosure, supplemented by a hardwired Emergency Stop (E-Stop) button for manual power isolation and an automated mechatronic protective system.
- **ISO 13857:2008 (Safety of Machinery — Safety distances)**: Dictates the physical dimensions of feeding chutes and barrier placements to prevent human limbs from reaching hazardous moving parts. Because the machine utilizes a bulk-load hopper rather than human hand-feeding, the physical distance through the hopper down to the roller intake acts as an absolute protective barrier. This distance ensures that mechanical nip-points remain physically inaccessible during normal operation.

2. Electrical & Electronics Standards

- **IEC 60204-1 (Safety of Machinery — Electrical Equipment of Machines):** Governs the electrical distribution path from the power supply unit (PSU) to the microcontroller and the high-torque gear motor. It ensures proper insulation, grounding of the

Aluminum T-slot chassis, and electrical isolation of low-voltage control signals from raw power lines to prevent short circuits or shocks.

- **IEEE 802.11 / Bluetooth SIG Standards:** the integrated microcontroller or control system utilizes wireless modules (ESP32 Wi-Fi/BLE protocols) for IoT status monitoring, the communication firmware complies with these standards to prevent electromagnetic interference (EMI) with neighboring laboratory or industrial equipment.

3. Food and Agricultural Material Standards

- **ISO 21469:2006 (Safety of machinery — Lubricants with incidental product contact)**: Ensures that any grease or lubricants used on the roller bearings or drive gears are non-toxic and rated for agricultural processing environments.
- **FDA Title 21 CFR (Material Safety for Food Contact Surfaces):** While the physical laboratory prototype utilizes 3D-printed PLA and Acrylic for rapid prototyping and cost-efficiency, the commercial design path specifies upgrading the core contact surfaces to AISI 304 Stainless Steel and food-grade Nitrile Butadiene Rubber (NBR) to fully comply with global food-safety and corrosion-resistance standards.

Legal and Intellectual Property (IP) Implications

1. Freedom to Operate (FTO) & Prior Art Assessment

A preliminary prior art assessment and patent clearance search were conducted utilizing international databases, including the World Intellectual Property Organization (WIPO) PATENTSCOPE and the United States Patent and Trademark Office (USPTO). This comprehensive evaluation verified that the machine's specific mechanical layout, namely, an asymmetric drive system utilizing a single actively driven roller coupled to a passive, friction-driven idler for agricultural pod extraction does not infringe upon active utility patents held by commercial agricultural machinery manufacturers. Consequently, the project maintains a clear Freedom to Operate (FTO) pathway within the target small-scale agricultural market.

2. Intellectual Property (IP) Protection Strategy

The structural and mechatronic design of this system presents a distinct, multi-layered opportunity for intellectual property management, bifurcated into public-domain and proprietary layers:

- **The Public-Domain Layer (Commercial Off-The-Shelf Integration):** In alignment with the project's Design for Serviceability (DFS) mandates, the structural assembly integrates extensive commercial off-the-shelf (COTS) components. The utilization of standardized metric fasteners (such as M3, M4, and M5 hardware), standardized

structural framing (6061 Aluminum T-slot extrusions), and standard-framed DC gear motors relies entirely on existing, non-patentable public-domain technology.
- **The Proprietary Layer (Utility & Industrial Design Patentability):** Conversely, the novel configuration of the system's specialized sub-assemblies represents protectable intellectual property. Specifically, the unique physical geometry of the custom 3D-printed PLA internal housing, the geometric orientation of the extraction roller assembly, the timed servo-motor-driven bulk feeding mechanism, and the exact closed-loop control logic deployed for overcurrent jam detection constitute proprietary technical innovations. These integrated assemblies qualify for Utility Patent protection or Industrial Design Registration based on their unique mechanical arrangement, hardware-software synchronization, and functional utility in resolving small-scale agricultural processing bottlenecks.

Planned Commercialization Framework: Strategic Licensing and Deployment

To maximize the regional impact of our automated bean pod extraction machine, we are structuring this project around a phased commercialization strategy that transitions from localized proprietary protection to an open-source deployment model:

- **Phase 1: Proprietary IP Protection and Manufacturing Licensing:** The initial phase of our deployment focuses on securing our proprietary mechatronic components, custom physical geometries, and control algorithms via restrictive utility licensing. We plan to pitch this protected architecture to local commercial agricultural equipment manufacturers. By establishing a localized proprietary licensing agreement, we aim to secure the necessary industrial backing to scale our production from 3D-printed PLA prototypes to heavy-duty, food-safe stainless steel commercial units.
- **Phase 2: Open-Source Prototyping and Community Deployment:** In parallel with commercial manufacturing, we plan to deploy an open-source initiative targeted at small-scale local farmers, agricultural cooperatives, and grassroots researchers. We intend to publish our hardware designs and firmware under an open-source framework, specifically utilizing the Creative Commons (CC BY-NC) license for physical assets and the GNU General Public License (GPL) for the control software.

Through this open-source pathway, we will make our digital design repository including our 3D-printed PLA component files (STL/STEP formats) and Bill of Materials (BOM) registries freely available for download. This strategy will empower end-users to independently construct or repair the machine utilizing locally sourced aluminum extrusions and standard commercial off-the-shelf (COTS) hardware. By actively executing this dual-layered framework, we will directly bridge the identified regional market gap, successfully lowering the capital expenditure barrier that currently prevents small-scale agricultural enterprises from adopting automation.

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Appendix A - coding

```
/*
 * ================================================================
 * PROJECT MAIN CONTROLLER — PRODUCTION VERSION (V9.0)
 * ================================================================
 * - WiFi SSID: Qash (Background connection)
 * - Roller Drive: FULLY ACTIVE ⚙️
 * - Servo Timing: STRICT HARDWARE LOCK TO PREVENT OVERSPIN 🔒
 * - Remote Control 1: MODERNISED LOCAL WEB DASHBOARD ACTIVE 🌐🎨
 * 👉 Local URL: http://extractionhub.local
 * - Remote Control 2: DUAL-USER TELEGRAM EMERGENCY ALERTS ACTIVE 📱📱
 * - Panel Interface: PIN 22 HARDWARE INTERRUPT & PIN 23 SMART ALERT LAMP
 * - Safety Guard: 2.9A Precision Instatrip Guard Engaged 🛡️
 * ================================================================
 */

#include <WiFi.h>
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>
#include <ArduinoJson.h>
#include <WebServer.h>
#include <ESPmDNS.h>

// ── Wi-Fi Credentials ────────────────────────────────────────────
#define WIFI_SSID     "Qash"
#define WIFI_PASSWORD "Qaisya_040411"

// ── Telegram Credentials ─────────────────────────────────────────
#define BOT_TOKEN      "8948272032:AAHnkcsOQ6CXLs0eC2X-LWNafpooVGi07cw"
#define CHAT_ID_1      "5393650784"   // Your Chat ID
#define CHAT_ID_2      "1034598864"   // Friend's Chat ID

// ── Pin Definitions ──────────────────────────────────────────────
#define MOTOR_A_IN1   25
#define MOTOR_A_IN2   26
#define MOTOR_A_ENA   32   // Conveyor Speed Control

#define VIB_IN3       27
#define VIB_IN4       14
#define VIB_ENB       33   // Vibrator Speed Control

#define MOTOR_B_IN1   16
#define MOTOR_B_IN2   17
#define MOTOR_B_ENA   18   // Roller Speed Control

#define RGB_R         4
```

```
#define RGB_G        5
#define RGB_B        19

#define SERVO_PIN     13   // 360-Degree Servo Pin
#define CURRENT_PIN   36   // Current Sensor Input (Labeled "VP")
#define RESET_PIN     22   // Push Button Switch Pin (Internal Pull-Up)
#define BUTTON_LED    23   // Push Button Built-In Red LED Positive (+) Pin

// ── PWM Settings ──────────────────────────────────────────
#define MOTOR_FREQ    5000
#define PWM_RES       8

#define SERVO_FREQ    50
#define SERVO_RES     8

// ── Speed Constants (0 to 255) ────────────────────────────
#define CONVEYOR_SPEED      255
#define ROLLER_SPEED        200   // Roller is ACTIVE
#define VIB_HALF_SPEED      128

// ── 360-SERVO TIMING LIMITS ───────────────────────────────
const int SERVO_REV_SPEED  = 13;
const int SERVO_STOP_SPEED = 19;
const int SERVO_FWD_SPEED  = 26;

// ── ACS712 Calibration Constants ──────────────────────────
const float ADC_VREF = 3.3;
const float ADC_RES = 4095.0;
const float SENSITIVITY = 0.100;
const float ACS712_RAW_ZERO_V = 2.50;
const float VOLTAGE_DIVIDER_RATIO = 3.0;

// ── Safety & State Settings ───────────────────────────────
const float CURRENT_SPIKE_THRESHOLD = 2.9;
bool systemFault = false;
bool bootMessageSent = false;
String lockdownReason = "None";

volatile bool physicalResetRequested = false;

String keyboardJson = "[[\"/start\", \"/stop\"], [\"/status\"]]";

unsigned long lastSensorRead = 0;
const unsigned long readInterval = 200;

unsigned long lastServoUpdate = 0;
unsigned long servoActionInterval = 500;
int servoState = 0;
```

```
unsigned long lastTelegramCheck = 0;
const unsigned long telegramInterval = 1000;

unsigned long systemStartTime = 0;
const unsigned long SENSOR_BYPASS_TIME = 4500;

// Initialize Client and Server Links
WiFiClientSecure secured_client;
UniversalTelegramBot bot(BOT_TOKEN, secured_client);
WebServer server(80);

void startSystem();
void stopSystem(float trippedCurrent, bool manualTrigger = false, String reason = "", String
origin = "AUTOMATIC");
void setRGB(uint8_t r, uint8_t g, uint8_t b);
float readCurrent();
void updateServo360TimeBased();
void checkResetButton();
void handleIncomingTelegramMessages(int numNewMessages);
void handleRootPage();
void handleStartCommand();
void handleStopCommand();

void IRAM_ATTR resetButtonISR() {
  if (digitalRead(RESET_PIN) == LOW) {
    physicalResetRequested = true;
  }
}

//
────────────────────────────────────────────────────────────────────────
// Setup
//
────────────────────────────────────────────────────────────────────────
void setup() {
  Serial.begin(115200);
  Serial.println("\n--- [PRODUCTION DEPLOYMENT V9.0 — HYBRID CORE RUNNING]
---");

  analogSetAttenuation(ADC_11db);

  pinMode(MOTOR_A_IN1, OUTPUT);
  pinMode(MOTOR_A_IN2, OUTPUT);
  pinMode(VIB_IN3,     OUTPUT);
  pinMode(VIB_IN4,     OUTPUT);
```

```
pinMode(MOTOR_B_IN1, OUTPUT);
pinMode(MOTOR_B_IN2, OUTPUT);
pinMode(CURRENT_PIN, INPUT);

pinMode(RESET_PIN,   INPUT_PULLUP);
pinMode(BUTTON_LED,  OUTPUT);

attachInterrupt(digitalPinToInterrupt(RESET_PIN), resetButtonISR, FALLING);

ledcAttach(RGB_R, MOTOR_FREQ, PWM_RES);
ledcAttach(RGB_G, MOTOR_FREQ, PWM_RES);
ledcAttach(RGB_B, MOTOR_FREQ, PWM_RES);

ledcAttach(MOTOR_A_ENA, MOTOR_FREQ, PWM_RES);
ledcAttach(VIB_ENB,     MOTOR_FREQ, PWM_RES);
ledcAttach(MOTOR_B_ENA, MOTOR_FREQ, PWM_RES);

ledcAttach(SERVO_PIN, SERVO_FREQ, SERVO_RES);
ledcWrite(SERVO_PIN, SERVO_STOP_SPEED);

startSystem();
systemStartTime = millis();

// Connect to Local Network
WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
Serial.print("[WIFI] Linking to Network: ");
Serial.println(WIFI_SSID);

while (WiFi.status() != WL_CONNECTED) {
  delay(500);
  Serial.print(".");
}

secured_client.setInsecure();

// Initialize mDNS Local Domain Routing
if (MDNS.begin("extractionhub")) {
  Serial.println("[mDNS] Local URL domain activated.");
}

// Clean layout printed once so logs do not slide away
Serial.println("\n==================================================");
Serial.println("[WIFI] Connected Successfully!");
Serial.print("[SERVER] Target IP Address: ");
Serial.println(WiFi.localIP());
Serial.println("[SERVER] Text Link Dashboard Setup Active 👇");
Serial.println("👉 URL: http://extractionhub.local");
Serial.println("==================================================\n");
```

```
  // Dashboard Control Bindings
  server.on("/", handleRootPage);
  server.on("/start", handleStartCommand);
  server.on("/stop", handleStopCommand);
  server.begin();
}

// ────────────────────────────────────────────────────────────────────────
// Main Loop
// ────────────────────────────────────────────────────────────────────────
void loop() {
  // Always service local browser clients instantly
  server.handleClient();

  // 1. Broadcast Cloud Startup Messages
  if (WiFi.status() == WL_CONNECTED && !bootMessageSent) {
    String bootMsg = "✅ *SYSTEM ONLINE (HYBRID NETWORK ACTIVE)*\n\n";
    bootMsg += "Extraction Hub is armed and broadcasting live telemetry.\n";
    bootMsg += "• *Local Panel:* http://extractionhub.local\n";
    bootMsg += "• *Safety Guard:* Enabled (2.9A Cap)";

    bot.sendMessageWithReplyKeyboard(CHAT_ID_1, bootMsg, "Markdown", keyboardJson, true);
    bot.sendMessageWithReplyKeyboard(CHAT_ID_2, bootMsg, "Markdown", keyboardJson, true);
    bootMessageSent = true;
  }

  // 2. Scheduled Time Slicing for Telegram Comm Links
  if (millis() - lastTelegramCheck >= telegramInterval) {
    lastTelegramCheck = millis();
    if (WiFi.status() == WL_CONNECTED) {
      int numNewMessages = bot.getUpdates(bot.last_message_received + 1);
      while (numNewMessages) {
        handleIncomingTelegramMessages(numNewMessages);
        numNewMessages = bot.getUpdates(bot.last_message_received + 1);
      }
    }
  }

  if (systemFault) {
    checkResetButton();
    return;
```

```
  }

  updateServo360TimeBased();

  // 3. Background Current Telemetry Loop
  if (millis() - lastSensorRead >= readInterval) {
    lastSensorRead = millis();
    float current = readCurrent();

    if (current < 0.0) {
      setRGB(0, 255, 0);
    }
    else {
      if (millis() - systemStartTime >= SENSOR_BYPASS_TIME) {
        if (current >= CURRENT_SPIKE_THRESHOLD) {
          stopSystem(current, false, "Automatic Overcurrent Jam Detection", "AUTOMATIC");
        }
      }
    }
  }
}

// ──────────────────────────────────────────────────────────────
// Telegram Messaging Loop Parsing Engine
// ──────────────────────────────────────────────────────────────
void handleIncomingTelegramMessages(int numNewMessages) {
  for (int i = 0; i < numNewMessages; i++) {
    String chat_id = String(bot.messages[i].chat_id);

    if (chat_id != CHAT_ID_1 && chat_id != CHAT_ID_2) {
      bot.sendMessage(chat_id, "Unauthorized device access denied.", "");
      continue;
    }

    String text = bot.messages[i].text;

    if (text == "/stop") {
      if (!systemFault) {
        stopSystem(readCurrent(), true, "Manual Cloud Override Request", "TELEGRAM");
      } else {
        bot.sendMessageWithReplyKeyboard(chat_id, "Machine is already in shutdown state.",
"", keyboardJson, true);
      }
    }
```

```
    else if (text == "/start") {
      if (systemFault) {
        physicalResetRequested = false;
        systemStartTime = millis();
        startSystem();

        bot.sendMessageWithReplyKeyboard(CHAT_ID_1, "▶️ *System Restored:* Control
panel cleared via Telegram.", "Markdown", keyboardJson, true);
        bot.sendMessageWithReplyKeyboard(CHAT_ID_2, "▶️ *System Restored:* Control
panel cleared via Telegram.", "Markdown", keyboardJson, true);
      } else {
        bot.sendMessageWithReplyKeyboard(chat_id, "Machine is already running normally.",
"", keyboardJson, true);
      }
    }
    else if (text == "/status") {
      float currentAmp = readCurrent();
      String currentStr = (currentAmp < 0.0) ? "0.00 (Offline)" : String(currentAmp, 2) + " A";

      String report = "📊 *LIVE CLOUD TELEMETRY REPORT*\n\n";
      report += "• *Machine Mode:* " + String(systemFault ? "🔴 LOCKED DOWN" : "🟢
ACTIVE RUNNING") + "\n";
      report += "• *Current Load:* " + currentStr + "\n";
      report += "• *Local Address:* http://extractionhub.local\n";
      report += "• *Roller Unit:* ACTIVE 🟢";

      bot.sendMessageWithReplyKeyboard(chat_id, report, "Markdown", keyboardJson, true);
    }
  }
}

//
────────────────────────────────────────────────────────────
─────
//  Industrial UI Local Dashboard Markup Engine
//
────────────────────────────────────────────────────────────
─────
void handleRootPage() {
  float currentAmp = readCurrent();
  String currentStr = (currentAmp < 0.0) ? "0.00 (Offline)" : String(currentAmp, 2) + " A";

  String html = "<!DOCTYPE html><html lang='en'><head>";
  html += "<meta charset='UTF-8'>";
  html += "<meta name='viewport' content='width=device-width, initial-scale=1.0'>";
  html += "<meta http-equiv='refresh' content='2'>";
  html += "<title>Mechatronics Extraction Hub</title>";
  html += "<style>";
```

```
  html += "body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
background-color: #0e1621; color: #f5f6f7; margin: 0; padding: 20px; display: flex;
justify-content: center; }";
  html += ".container { width: 100%; max-width: 650px; margin-top: 10px; }";
  html += "header { text-align: center; margin-bottom: 25px; border-bottom: 2px solid
#1b2836; padding-bottom: 15px; }";
  html += "header h1 { margin: 0; font-size: 26px; color: #4fc3f7; letter-spacing: 1px; }";
  html += "header p { margin: 5px 0 0 0; color: #8b9bb4; font-size: 13px; text-transform:
uppercase; letter-spacing: 2px; }";

  if (systemFault) {
    html += ".main-status { background: #2c1a1d; border: 2px solid #e53935; border-radius:
12px; padding: 20px; text-align: center; font-size: 20px; font-weight: bold; color: #ff5252;
box-shadow: 0 0 15px rgba(229,57,53,0.2); margin-bottom: 20px; }";
  } else {
    html += ".main-status { background: #16261b; border: 2px solid #43a047; border-radius:
12px; padding: 20px; text-align: center; font-size: 20px; font-weight: bold; color: #69f0ae;
box-shadow: 0 0 15px rgba(67,160,71,0.2); margin-bottom: 20px; }";
  }

  html += ".grid { display: grid; grid-template-columns: 1fr 1fr; gap: 15px; margin-bottom:
25px; }";
  html += ".tile { background: #17212b; border: 1px solid #243242; border-radius: 10px;
padding: 15px 20px; box-shadow: 0 4px 6px rgba(0,0,0,0.15); }";
  html += ".tile-title { font-size: 11px; color: #7f8c8d; text-transform: uppercase; font-weight:
600; letter-spacing: 1px; margin-bottom: 5px; }";
  html += ".tile-value { font-size: 18px; font-weight: bold; color: #ffffff; }";

  html += ".panel-actions { display: flex; gap: 15px; }";
  html += ".btn { flex: 1; padding: 18px; font-size: 16px; font-weight: bold; border: none;
border-radius: 8px; cursor: pointer; text-decoration: none; text-align: center; transition: all
0.2s ease; box-shadow: 0 4px 10px rgba(0,0,0,0.3); }";
  html += ".btn-start { background-color: #00e676; color: #0a0f14; }";
  html += ".btn-start:hover { background-color: #69f0ae; transform: translateY(-1px); }";
  html += ".btn-stop { background-color: #ff1744; color: #ffffff; }";
  html += ".btn-stop:hover { background-color: #ff5252; transform: translateY(-1px); }";
  html += ".btn:active { transform: translateY(1px); box-shadow: 0 2px 5px rgba(0,0,0,0.2); }";
  html += "</style></head><body>";

  html += "<div class='container'>";
  html += "<header>";
  html += "<h1>PODCRUSH AUTOMATIC</h1>";
  html += "<p>[ AUTOMATIC BEAN POD EXTRACTION MACHINE ]</p>";
  html += "</header>";

  if (systemFault) {
    html += "<div class='main-status'>⚠️ EMERGENCY SYSTEM LOCKDOWN
ACTIVE</div>";
```

```
  } else {
    html += "<div class='main-status'>⚙️ ALL PLANT ACTUATORS OPERATIONAL</div>";
  }

  html += "<div class='grid'>";

  html += "<div class='tile'>";
  html += "<div class='tile-title'>Live Telemetry Load</div>";
  html += "<div class='tile-value' style='color:#4fc3f7;'>" + currentStr + "</div>";
  html += "</div>";

  html += "<div class='tile'>";
  html += "<div class='tile-title'>Roller Unit Driver</div>";
  html += "<div class='tile-value' style='color:" + String(systemFault ? "#ff5252" : "#69f0ae") +
";'>" + String(systemFault ? "OFF" : "ACTIVE (200)") + "</div>";
  html += "</div>";

  html += "<div class='tile' style='grid-column: span 2;'>";
  html += "<div class='tile-title'>Active Mode Diagnostic Description</div>";
  html += "<div class='tile-value' style='font-size:15px; font-weight:normal; color:#b0bec5;'>"
+ lockdownReason + "</div>";
  html += "</div>";

  html += "</div>";

  html += "<div class='panel-actions'>";
  html += "<a href='/start' class='btn btn-start'>START SYSTEM</a>";
  html += "<a href='/stop' class='btn btn-stop'>STOP SYSTEM</a>";
  html += "</div>";

  html += "</div></body></html>";

  server.send(200, "text/html", html);
}

void handleStartCommand() {
  if (systemFault) {
    Serial.println("[SERVER] Remote startup signal received via local browser link.");
    physicalResetRequested = false;
    startSystem();

    bot.sendMessageWithReplyKeyboard(CHAT_ID_1, "▶️ *System Restored:* Cleared via
local Web Dashboard.", "Markdown", keyboardJson, true);
    bot.sendMessageWithReplyKeyboard(CHAT_ID_2, "▶️ *System Restored:* Cleared via
local Web Dashboard.", "Markdown", keyboardJson, true);
  }
  server.sendHeader("Location", "/");
  server.send(303);
```

```
}

void handleStopCommand() {
  if (!systemFault) {
    Serial.println("[SERVER] Remote emergency stop triggered by operator via web client.");
    stopSystem(readCurrent(), true, "Manual Web Dashboard Shutdown Override",
"WEB_DASHBOARD");
  }
  server.sendHeader("Location", "/");
  server.send(303);
}

// ────────────────────────────────────────────
───
//  Reset Button Handler (Interrupt Driven)
// ────────────────────────────────────────────
───
void checkResetButton() {
  if (physicalResetRequested) {
    delay(50);

    Serial.println("\n[RESET] Physical Maintenance Button Pressed on Control Panel!");
    physicalResetRequested = false;
    startSystem();

    bot.sendMessageWithReplyKeyboard(CHAT_ID_1, "▶️ *System Restored:* Hardware
panel reset completed.", "Markdown", keyboardJson, true);
    bot.sendMessageWithReplyKeyboard(CHAT_ID_2, "▶️ *System Restored:* Hardware
panel reset completed.", "Markdown", keyboardJson, true);

    while(digitalRead(RESET_PIN) == LOW) { delay(10); }
  }
}

// ────────────────────────────────────────────
───
//  System State Control Helpers
// ────────────────────────────────────────────
───
void startSystem() {
  systemFault = false;
  lockdownReason = "Running standard automation routines safely.";
  setRGB(0, 255, 0);
  digitalWrite(BUTTON_LED, LOW);
```

```
  digitalWrite(MOTOR_A_IN1, LOW);
  digitalWrite(MOTOR_A_IN2, HIGH);
  ledcWrite(MOTOR_A_ENA, CONVEYOR_SPEED);

  digitalWrite(VIB_IN3, HIGH);
  digitalWrite(VIB_IN4, LOW);
  ledcWrite(VIB_ENB, VIB_HALF_SPEED);

  digitalWrite(MOTOR_B_IN1, LOW);
  digitalWrite(MOTOR_B_IN2, HIGH);
  ledcWrite(MOTOR_B_ENA, ROLLER_SPEED);

  Serial.println("[SYSTEM ONLINE] Actuators and extraction sub-units active.");
}

void stopSystem(float trippedCurrent, bool manualTrigger, String reason, String origin) {
  systemFault = true;
  if (manualTrigger) {
    lockdownReason = reason;
  } else {
    lockdownReason = reason + " (Tripped at: " + String(trippedCurrent, 2) + "A)";
  }
  setRGB(255, 0, 0);
  digitalWrite(BUTTON_LED, HIGH);

  digitalWrite(MOTOR_A_IN1, LOW);
  digitalWrite(MOTOR_A_IN2, LOW);
  ledcWrite(MOTOR_A_ENA, 0);

  digitalWrite(VIB_IN3, LOW);
  digitalWrite(VIB_IN4, LOW);
  ledcWrite(VIB_ENB, 0);

  digitalWrite(MOTOR_B_IN1, LOW);
  digitalWrite(MOTOR_B_IN2, LOW);
  ledcWrite(MOTOR_B_ENA, 0);

  ledcWrite(SERVO_PIN, SERVO_STOP_SPEED);

  Serial.println("\n[!!!!] LOCKDOWN EXECUTED.");

  // Broadcast the event to the cloud layer immediately
  String msg = "⚠️ *SYSTEM LOCKDOWN EXECUTED* ⚠️\n\n";
  msg += "• *Type:* " + origin + "\n";
  msg += "• *Description:* " + reason + "\n";
  msg += "• *Current Load:* " + String(trippedCurrent, 2) + " A\n\n";
  msg += "Reset locally via web link or hit the physical switch.";
```

```
  bot.sendMessageWithReplyKeyboard(CHAT_ID_1, msg, "Markdown", keyboardJson, true);
  bot.sendMessageWithReplyKeyboard(CHAT_ID_2, msg, "Markdown", keyboardJson, true);
}

// ── STRICT HARDWARE LOCK SERVO CONTROLLER ──────────────────────────
void updateServo360TimeBased() {
  if (millis() - lastServoUpdate >= servoActionInterval) {
    if (servoState == 0) {
      ledcWrite(SERVO_PIN, SERVO_FWD_SPEED);
      delay(180);
      ledcWrite(SERVO_PIN, SERVO_STOP_SPEED);

      servoActionInterval = 800;
      lastServoUpdate = millis();
      servoState = 1;
    }
    else if (servoState == 1) {
      ledcWrite(SERVO_PIN, SERVO_REV_SPEED);
      delay(180);
      ledcWrite(SERVO_PIN, SERVO_STOP_SPEED);

      servoActionInterval = 5000;
      lastServoUpdate = millis();
      servoState = 0;
    }
  }
}

void setRGB(uint8_t r, uint8_t g, uint8_t b) {
  ledcWrite(RGB_R, r);
  ledcWrite(RGB_G, g);
  ledcWrite(RGB_B, b);
}

float readCurrent() {
  long adcRawSum = 0;
  const int samples = 20;

  for (int i = 0; i < samples; i++) {
    adcRawSum += analogRead(CURRENT_PIN);
  }

  float adcAverage = (float)adcRawSum / samples;

  if (adcAverage < 150.0) {
    return -1.0;
  }
```

```
  float pinVoltage = (adcAverage / ADC_RES) * ADC_VREF;
  float sensorVoltageFullScale = pinVoltage * VOLTAGE_DIVIDER_RATIO;
  float current = (sensorVoltageFullScale - ACS712_RAW_ZERO_V) / SENSITIVITY;

  if (abs(current) < 0.15) {
    current = 0.0;
  }

  return abs(current);
}
```

APPENDIX B BOM

Item Description	Sourcing Type	Quantity	Unit / Lot Price
Custom 3D-Printed Feeder Mechanism (PLA)	Custom Fabricated	1 Lot	112.72
Power Box Enclosure Units	Standard COTS	2 Units	60.00
Laser-Cut Transparent Acrylic Safety Guards	Custom Fabricated	1 Lot	46.19
Conveyor Assembly Core	Standard COTS	1 Unit	29.00
Custom 3D-Printed Transmission Gears (PLA)	Custom Fabricated	1 Lot	24.32
High-Friction Rubber Conveyor Belt Tape	Standard COTS	1 Roll	19.76

Deep-Groove Ball Bearings	Standard COTS	4 Units	17.60
Calibration Weight Hardware	Standard COTS	1 Unit	9.81
Custom 3D-Printed Conveyor Cylinder (PLA)	Custom Fabricated	1 Unit	3.45
Mechanical Subtotal			**322.85**

TABLE BOM FOR MECHANICAL PART

Item Description	**Sourcing Type**	**Quantity**	**Unit / Lot Price**
Primary Inline Overcurrent Current Sensor	Standard COTS	1 Unit	18.99
MG996R High-Torque Dispensing Servo Motor	Standard COTS	1 Unit	15.00
DC Agitation Vibrator Motor (Feeder Hopper)	Standard COTS	1 Unit	14.00
20AWG Stranded Core Electrical Cabling	Standard COTS	1 Lot	13.89
DC-DC Step-Down Buck Converter Module	Standard COTS	1 Unit	12.39

High-Current T-Plug Dean Connectors	Standard COTS	1 Lot	10.00
Female-to-Male Jumper Wire Harnesses	Standard COTS	1 Lot	8.87
Female-to-Female Jumper Wire Harnesses	Standard COTS	1 Lot	7.78
Electrical Subtotal			**100.92**

TABLE BOM FOR ELECTRICAL PART

Item Description	**Sourcing Type**	**Quantity**	**Unit / Lot Price**
Standard M6 Hex Metric Screws	Standard COTS	18 Units	37.10
Modular Corner Slot Panel Joiners	Standard COTS	6 Units	23.22
Standard M5 Hex Metric Screws	Standard COTS	18 Units	20.00
Threaded Brass Hex Spacers	Standard COTS	30 Units	19.24

Ball Spring T-Slot Nuts (2020 Profile)	Standard COTS	20 Units	14.79
T-Slot Modular L-Brackets (M5 Profile)	Standard COTS	10 Units	13.20
T-Slot Structural Aluminum Extrusions (2020 M4)	Standard COTS	10 Units	10.00
Standard M5 Button-Head Screws	Standard COTS	20 Units	9.39
Standard M4 Button-Head Screws	Standard COTS	20 Units	8.39
T-Slot Slide-In Nuts (M5 Profile)	Standard COTS	10 Units	8.10
Standard M3 Pan-Head Machine Screws	Standard COTS	30 Units	6.82
Flexible Motor Shaft Couplers	Standard COTS	2 Units	5.45
Fasteners & Structural Subtotal			**175.70**

TABLE BOM FOR Fasteners & Structural PART

Item Description	Sourcing Type	Quantity	Unit / Lot Price
Raw Edamame Feedstock (Testing & Validation Cycles)	Consumable	1 Lot	81.00
Electrical Wire Ferrule Crimp Connectors	Consumable	1 Box	15.59
Machine Threading Taps	Tooling	2 Units	12.26
Miscellaneous Subtotal			**108.85**

TABLE BOM FOR MISCELLANEOUS PART

Project Subsystem Module	Allocated Cost	Relative Budget Share (%)
Mechanical Components & Custom Fabrication	322.85	45.6%
Fasteners, Couplers & Structural Hardware	175.70	24.8%
Operational Consumables & Validation Feedstock	108.85	15.4%

Electrical Components, Wiring & Actuators	100.92	14.2%
Grand Total Project Expenditure	**708.32**	**100.0%**